

WireGuard Core (helix-wg)

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Volume 2, document 01 of the HelixVPN master technical specification — the nano-detail deepening of the **WireGuard Core (helix-wg)** section of the pass-1 data-plane overview [01-DP §4]. This is a SPEC: it describes *what to build* (every trait/struct/enum/fn signature, wire layout, state machine, error taxonomy, config knob, edge case, security note, performance budget, and test point) — it does not ship the product. Sources are cited inline by id.

Evidence-base honesty notes (constitution §11.4.6 — stated, not guessed)

Three grounding caveats apply to this whole document; every claim that rests on them carries an inline UNVERIFIED: marker:

1. **research-wireguard.md does not exist** in the supplied evidence base (scratchpad/kb/ contains research-mullvad.md, research-masque.md, research-hysteria2.md, ... but no research-wireguard.md). Therefore every **WireGuard protocol wire-byte layout** (handshake message sizes, field offsets, MAC layout, transport-data header) in §3 is UNVERIFIED: against the evidence base — it is reproduced from the public WireGuard protocol (Donenfeld, *WireGuard: Next Generation Kernel Network Tunnel*) and MUST be re-confirmed against the WireGuard whitepaper + RFC-equivalent + the boringtun source before implementation. The *architectural* facts (Noise IK, Curve25519, ChaCha20-Poly1305, BLAKE2s, PSK seam) ARE grounded [01-DP §4, SYNTHESIS §2, research-mullvad §4].
2. **boringtun internal API signatures** (Tunn::encapsulate, decapsulate, update_timers, TunnResult) are UNVERIFIED: against the evidence base — they are reproduced from the public boringtun crate (cloudflare/boringtun) and MUST be pinned to the exact crate version's API at

implementation time. The evidence base grounds only the *wrapper* shape (WgPeer, WgVerdict, the three pump methods) [01-DP §4, 04_P0 §4.4].

3. §11.4.169 ("**test types**") referenced in the authoring brief is **not present** in this document's constitution context (which ends at §11.4.168). UNVERIFIED: its exact enumeration. The concrete test types in §13 are therefore drawn from the in-context §11.4.27 test-type set (unit, integration, e2e, full-automation, security, fuzz, stress, chaos, performance, benchmark, Challenges, HelixQA) + the §11.4.5/.69/.107 captured-evidence covenant, and labelled "§11.4.169-class" where the brief intends the per-component mandate.
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0. Position, ownership, and invariants

`helix-wg` is the crate at **L3 — the WireGuard cryptographic core** of the layering model [01-DP §1]. It is the *only* crate in the workspace that touches WireGuard plaintext on one side and WireGuard ciphertext on the other; the Transport layer (§3 of [01-DP]) sits strictly **beneath** it and never sees plaintext (invariant I1) [01-DP §0.1]. `helix-wg` owns the WireGuard Tunn state machine, pumps its timers, routes its output verdicts, and enforces per-peer AllowedIPs crypto-routing. It is the same crate byte-for-byte on client, connector, and gateway edge (invariant I4) [01-DP §0.1, §4; 04_ARCH §3.2].

What `helix-wg` owns (this document):

- boringtun-vs-kernel selection per platform (§1).
- The WgPeer / WgDevice wrapper over boringtun Tunn, its WgVerdict output enum, and the four pump entry points (§2).
- The Noise IK handshake state machine + WireGuard message wire formats it drives (§3) — UNVERIFIED: byte layouts per note 1.
- The fixed crypto parameter set + the additive PSK (post-quantum) seam (§4).
- AllowedIPs semantics: the longest-prefix-match crypto-routing trie (§5).
- Per-device key generation + rotation lifecycle (§6).
- MTU derivation from the active transport's `effective_mtu()` (§7).
- Timer / rekey / keepalive state machine (§8).
- The encrypted-datagram boundary contract with `helix-transport` (§9).
- Error taxonomy (§10), config knobs (§11), edge cases (§12), test points (§13), security (§14), performance budget (§15).

What `helix-wg` does NOT own: the obfuscating transport (`helix-transport`, [01-DP §3]); the TUN device (`helix-tun`); the orchestrator's three loops and TunnelStatus stream (`helix-core`, [01-DP §5]); the route

map / ACL compiler (helix-route, [01-DP §6–§7]); DAITA shaping (helix-daita, [01-DP §9]); the full post-quantum KEM exchange protocol (security doc — helix-wg exposes only the PSK *field*, §4.4).

0.1 Crate-local invariants (extend the §0.1 data-plane invariants)

#	Invariant	Source
W1	helix-wg performs zero heap allocation on the steady-state encrypt/decrypt path ; all framing uses caller-provided scratch buffers (&mut [u8]).	derived from iOS NE memory ceiling [04_P0 §6, 04_ARCH §5.6]
W2	Crypto parameters are fixed and never altered (Noise IK / Curve25519 / ChaCha20-Poly1305 / BLAKE2s); the only additive seam is the WG PSK field.	[01-DP §4, SYNTHESIS §2]
W3	A device private key is generated on-device and never leaves it ; helix-wg exposes only the public key for enrollment.	[01-DP §4, 04_ARCH §7, SYNTHESIS §7]
W4	A packet whose decrypted source / encrypted destination does not match the peer's AllowedIPs is dropped — crypto-routing is the coarse default-deny layer (I6).	[01-DP §7, 04_ARCH §3.4]
W5	helix-wg keeps no durable per-packet or per-connection state — only in-RAM session state + aggregate counters (no-logging by construction, I5).	[01-DP §0.1, 04_ARCH §2.7, research-mullvad §8]
W6	The wrapper is the same crate on all three roles ; role differences (client default-route capture vs connector CIDR forwarding) live in helix-core, not here.	[01-DP §4, §5.1]

1. boringtun (userspace) vs kernel WireGuard — per platform

1.1 The decision (grounded)

boringtun is the **default and the cross-platform floor**: pure-Rust userspace WireGuard, no kernel module, runs identically on Linux/Android/iOS — exactly the iOS path, where there is no kernel WG regardless [04_P0

§4.4]. On **Linux** the kernel WG fast path is used where available, with boringtun userspace as fallback; the project **ships both and defaults to kernel on Linux** [01-DP §4, 04_ARCH §13 "ship both, default to kernel"].

1.2 Per-platform matrix

Platform	Backend selected	Rationale	Source
Linux (client/ connector/ edge)	kernel WG (wireguard netlink/wg), boringtun fallback	kernel is faster; constrains containerization/ permissions → fallback exists	[04_ARCH §13, §3.2; 04_P0 §4.4]
Android	boringtun (via VpnService+JNI)	userspace, no root, uniform	[04_ARCH §5.3, SYNTHESIS §5]
iOS / macOS (NE)	boringtun (in NEPacketTunnelProvider)	no kernel WG on iOS; bounded- memory Rust is the make-or- break gate G3	[04_P0 §4.4, §6; 04_ARCH §5.6]
Windows	boringtun over wireguard-nt/wintun, or kernel-ish wireguard-nt data path	shim owns the adapter; backend abstracted	[04_ARCH §5.3]
HarmonyOS / Aurora	boringtun (native shim → .so)	no kernel WG; Phase 3	[04_ARCH §5.3, SYNTHESIS §4]

UNVERIFIED: exact Windows kernel-vs-userspace split — the evidence base says "wireguard-nt / wintun" [04_ARCH §5.3] without pinning whether boringtun or wireguard-nt's own crypto runs the data path; settle in the Windows shim spec (doc 06).

1.3 Backend abstraction — WgBackend

The two backends hide behind one trait so helix-core and the rest of the workspace are backend-agnostic; backend selection is a build-time + runtime choice resolved once at device construction.

```
// helix-wg/src/backend.rs
/// Selects how WireGuard crypto + the Tunn state machine are
    executed.
#[derive(Clone, Copy, Debug, PartialEq, Eq)]
```

```

pub enum WgBackend {
    /// Pure-Rust userspace (boringtun Tunn). The cross-platform
    /// floor.
    Userspace,
    /// Linux kernel WireGuard via netlink (`wg`/`wireguard`
    /// module).
    LinuxKernel,
}

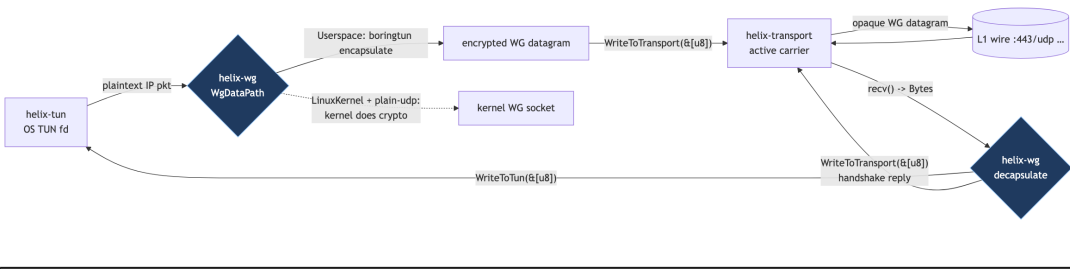
impl WgBackend {
    /// Resolve the backend for the current platform +
    /// availability.
    /// Linux: prefer LinuxKernel if the module is loadable AND
    /// not running
    /// inside a constrained container; else Userspace. All other
    /// platforms: Userspace.
    pub fn resolve(prefer_kernel: bool) -> WgBackend { /* ... */ }
}

/// The role-neutral handle the orchestrator drives, regardless of
/// backend.
/// (Userspace impl wraps boringtun Tunn per §2; LinuxKernel impl
/// drives netlink.)
pub trait WgDataPath: Send + Sync {
    /// Encrypt one outbound plaintext IP packet for `peer_index`
    /// (userspace path);
    /// kernel path is configured once and the kernel does this –
    /// see §9.3.
    fn encapsulate<'a>(&mut self, peer: PeerIndex, ip_pkt: &[u8],
        scratch: &'a mut [u8])
        -> WgVerdict<'a>;
    fn decapsulate<'a>(&mut self, datagram: &[u8], scratch: &'a
        mut [u8]) -> WgVerdict<'a>;
    fn tick<'a>(&mut self, scratch: &'a mut [u8]) ->
        WgVerdict<'a>;
    fn backend(&self) -> WgBackend;
}

```

Kernel-path note (§9.3 expands): when LinuxKernel is selected, the per-packet encrypt/decrypt happens *inside the kernel*, NOT through encapsulate/decapsulate. In that mode helix-wg (a) programs the kernel peer set + AllowedIPs via netlink, and (b) the *transport* must still be userspace for any non-plain-udp carrier — so a kernel/userspace **split** applies only to plain-udp; every obfuscating transport forces the userspace data path because the kernel cannot emit MASQUE/Shadowsocks framing. This is a first-class edge case (§12.7).

1.4 Backend placement in the packet flow



2. The helix-wg boringtun Tunn wrapper + WgVerdict

2.1 The four-verdict output model (grounded)

boringtun's Tunn is a sans-I/O state machine: you feed it bytes and it returns a verdict telling you what to do next [04_P0 §4.4]. helix-wg maps boringtun's TunnResult onto a tighter, lifetime-bound WgVerdict [01-DP §4]:

```
// helix-wg/src/lib.rs
/// The output of every helix-wg pump call. Borrows the caller's
///   scratch buffer
/// (W1: no heap alloc on the hot path) – the orchestrator acts on
///   it immediately
/// before the next pump call reuses the scratch.
pub enum WgVerdict<'a> {
    /// Encrypted datagram (handshake or transport-data) → hand to
    ///   the active Transport (§3 of 01-DP).
    WriteToTransport(&'a [u8]),
    /// Decrypted inner IP packet → hand to the TUN. Carries the
    ///   parsed inner src for AllowedIPs audit (§5).
    WriteToTun { pkt: &'a [u8], inner_src: IpAddr },
    /// Nothing to emit this call (e.g. duplicate, keepalive
    ///   consumed, queued).
    Nothing,
    /// A genuine WireGuard-layer error (see WgError, §10). NEVER
    ///   used for "no output".
    Err(WgError),
}
```

Mapping from boringtun TunnResult (UNVERIFIED: boringtun API per note 2; this is the public crate surface to confirm at pin time):

boringtun TunnResult	helix-wg WgVerdict	Notes
Done	Nothing	nothing to emit
Err(WireGuardError)	Err(WgError::from(_))	


```

    local_private: PrivKey,           // generated on-device, never
        leaves (W3)
    local_public: PubKey,
    peers: Vec<WgPeer>,
    routing: AllowedIpsTrie,          // §5 – maps inner src/dst →
        PeerIndex
    backend: WgBackend,
    rate_limiter: HandshakeRateLimiter, // §14.3 anti-DoS (cookie
        reply path)
    index_map: PeerIndexMap,          // WG receiver-index →
        PeerIndex demux (§3.5)
}

impl WgDevice {
    /// Construct from the local static key + initial peer set
    /// (from the RouteMap, 01-DP §6.2).
    pub fn new(local_private: PrivKey, backend: WgBackend) ->
        Self { /* ... */ }

    /// Add / update a peer (reconciler-driven, no restart – 01-DP
    /// §6.3). Idempotent on pubkey.
    pub fn upsert_peer(&mut self, cfg: PeerConfig) -> PeerIndex
        { /* ... */ }
    /// Remove a peer (e.g. device.revoked → drop in <1s – 01-DP
    /// §7). Zeroizes its session.
    pub fn remove_peer(&mut self, peer: PeerIndex) { /* ... */ }

    /// OUTBOUND: a plaintext IP packet left the TUN → find the
    /// peer by dst (§5),
    /// encrypt, return ciphertext for the transport. Drops (W4)
    /// if no AllowedIPs match.
    pub fn handle_tun_out<'a>(&mut self, ip_pkt: &[u8], scratch:
        &'a mut [u8]) -> WgVerdict<'a>;

    /// INBOUND: bytes arrived from the transport → demux to the
    /// right peer by WG
    /// receiver index (§3.5), decrypt; emit plaintext for the TUN
    /// (after §5 src audit)
    /// or a handshake reply for the transport.
    pub fn handle_transport_in<'a>(&mut self, datagram: &[u8],
        from: Option<SocketAddr>,
        scratch: &'a mut [u8]) ->
        WgVerdict<'a>;

    /// TIMER: call every TICK_INTERVAL (§8.1); emits keepalives,
    /// rekeys, expirations
    /// for ALL peers (round-robins which peer needs work this
    /// tick).
    pub fn tick<'a>(&mut self, now: Instant, scratch: &'a mut
        [u8]) -> Vec<WgVerdict<'a>>;

    pub fn local_public(&self) -> PubKey { self.local_public }
    pub fn peer_stats(&self, peer: PeerIndex) -> PeerStats { /*
        aggregate only, W5 */ }
}

```

```
#[derive(Clone, Debug, Default)]
pub struct PeerStats {           // W5: aggregate counters only,
    never per-packet logs
    pub tx_bytes: u64, pub rx_bytes: u64,
    pub tx_packets: u64, pub rx_packets: u64,
    pub handshakes_ok: u32, pub handshakes_failed: u32,
    pub last_handshake_age_s: Option<u32>,
}

pub struct PeerConfig {
    pub remote_pubkey: PubKey,
    pub allowed_ips: Vec<IpNet>,
    pub endpoint: Option<SocketAddr>,
    pub persistent_keepalive: Option<u16>,
    pub psk: Option<Psk>,         // §4.4
}
```

2.3 Why `handle_tun_out` takes the whole device, not a single peer

A multi-hop chain or an edge with N peers must select the *right* peer from the inner destination IP via the AllowedIPs trie (§5) before encrypting — so the outbound entry point lives on `WgDevice`, not `WgPeer`. The single-peer client case is just N=1. (The pass-1 sketch put `handle_tun_out` on `WgPeer` for the one-peer slice [01-DP §4]; this nano-spec lifts it to `WgDevice` for the edge's N-peer reality, a deliberate refinement — the one-peer body is unchanged.)

3. Noise IK handshake + WireGuard message wire formats

UNVERIFIED: (note 1) — the byte sizes/offsets below are the public WireGuard protocol values, NOT confirmed against an evidence-base `research-wireguard.md` (which is absent). Confirm against the WireGuard whitepaper + boringtun source before coding. The *architecture* (Noise IK, 1-RTT, the four message types, the MAC1/MAC2 cookie anti-DoS) is grounded [01-DP §4, research-mullvad §4 PQ-PSK-over-WG].

3.1 Handshake shape (grounded architecture)

WireGuard uses the **Noise IK** pattern: a **1-RTT** handshake where the initiator already knows the responder's static public key (the "I" = initiator immediately transmits its static key encrypted; "K" = responder's static key

is known in advance) [01-DP §4]. HelixVPN learns the responder's static pubkey from the RouteMap (PeerRoute.wg_pubkey) pushed by the coordinator [01-DP §6.2].

Four message types (UNVERIFIED: sizes):

Type	Name	Size (bytes)	Direction
1	Handshake Initiation	148	initiator → responder
2	Handshake Response	92	responder → initiator
3	Cookie Reply	64	responder → initiator (under load / anti-DoS, §14.3)
4	Transport Data	$16 + 16 \cdot \lceil (\text{plaintext} + 0) / 16 \rceil$... header 16 + ciphertext	both, steady state

3.2 Message 1 — Handshake Initiation (148 B, UNVERIFIED: layout)

offset	size	field	
0	1	message_type = 0x01	
1	3	reserved = 0x000000	
4	4	sender_index	(u32 LE, initiator's receiver index)
8	32	unencrypted_ephemeral public)	(Curve25519 ephemeral public)
40	48	encrypted_static	(32 B static pub + 16 B Poly1305 tag)
88	28	encrypted_timestamp	(12 B TAI64N + 16 B tag)
116	16	mac1	(BLAKE2s keyed MAC over preceding bytes)
132	16	mac2	(cookie MAC; zero unless under cookie challenge)

3.3 Message 2 — Handshake Response (92 B, UNVERIFIED: layout)

offset	size	field	
0	1	message_type = 0x02	
1	3	reserved = 0x000000	
4	4	sender_index	(responder's receiver index)
8	4	receiver_index	(echoes initiator's sender_index)
12	32	unencrypted_ephemeral	
44	16	encrypted_nothing	(empty payload + 16 B tag – confirms key schedule)

60	16	mac1
76	16	mac2

3.4 Message 4 — Transport Data header (UNVERIFIED: layout)

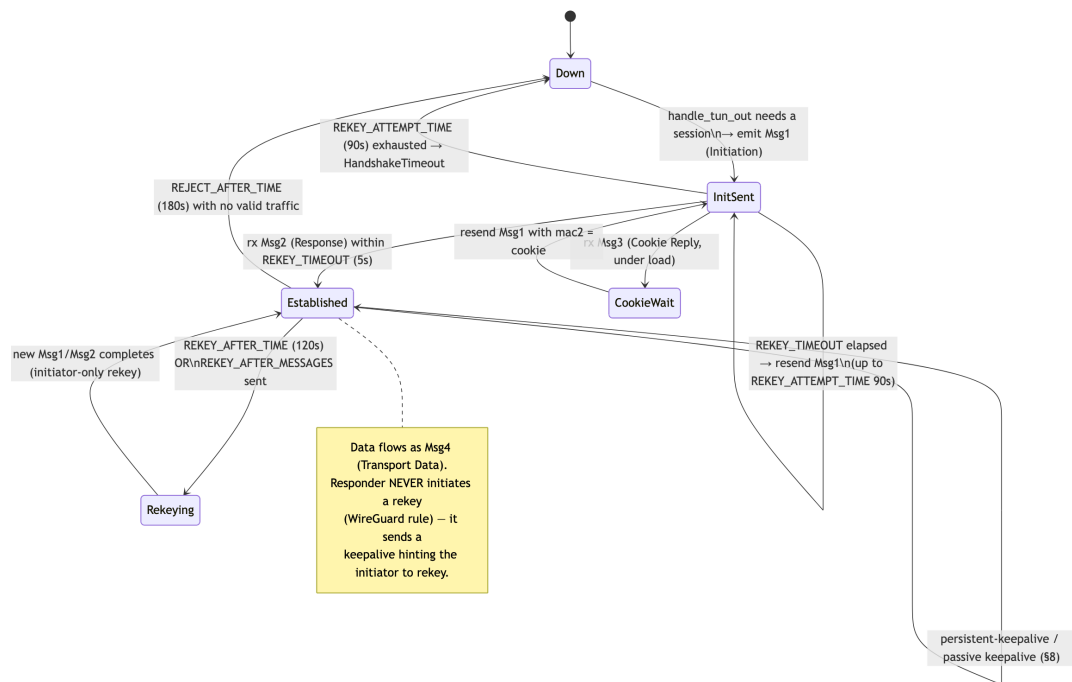
offset	size	field	
0	1	message_type = 0x04	
1	3	reserved = 0x000000	
4	4	receiver_index	(peer's receiver index – demux key, §3.5)
8	8	counter	(u64 LE nonce; monotonic per session, anti-replay)
16	N	encrypted_packet	(ChaCha20–Poly1305 of the inner IP packet, incl. 16 B tag)

The plaintext is padded to a multiple of 16 bytes before AEAD (WireGuard padding); the 16-byte Poly1305 tag is appended inside encrypted_packet. This fixed 16-byte data header + 16-byte tag is the **32-byte WG overhead** that the MTU budget (§7) accounts for.

3.5 Receiver-index demux (the edge's N-peer dispatch)

On inbound, handle_transport_in reads bytes [4..8) (the receiver_index, which the *local* device assigned at handshake) to select the PeerIndex via index_map before decrypting — O(1), no trial decryption across N peers. A datagram with an unknown receiver index, a type $\notin \{1,2,3,4\}$, or a malformed length is dropped as WgError::Malformed (§10), counted, not logged (W5).

3.6 Handshake state machine



Timer constants (UNVERIFIED: exact values — standard WireGuard defaults, pin against boringtun): REKEY_AFTER_TIME = 120s, REKEY_TIMEOUT = 5s, REKEY_ATTEMPT_TIME = 90s, REJECT_AFTER_TIME = 180s, KEYPALIVE_TIMEOUT = 10s, REKEY_AFTER_MESSAGES = 2⁶⁰, REJECT_AFTER_MESSAGES = 2⁶⁴ – 2¹³. These are surfaced as read-only constants, NOT config knobs (W2 — crypto invariants are fixed).

4. Crypto parameters (fixed) + the additive PSK seam

4.1 Fixed primitive set (grounded, W2)

Never altered [01-DP §4, SYNTHESIS §2]:

Role	Primitive
Key agreement	Curve25519 ECDH (X25519)
AEAD (data + handshake)	ChaCha20-Poly1305 (IETF, 96-bit nonce = 32-bit zero 64-bit counter)
Hash / KDF / MAC	BLAKE2s (HKDF-style key schedule; keyed BLAKE2s for MAC1/MAC2)
Handshake	Noise IK (1-RTT, known responder static key) TAI64N (12 B)

Role	Primitive
Timestamp (anti-replay of Msg1)	

4.2 Key schedule (grounded architecture; UNVERIFIED: exact chaining)

The Noise IK key schedule mixes: responder static pub, initiator ephemeral, initiator static, the optional PSK (§4.4), and BLAKE2s-chained Ck/Hk values to derive the two directional transport keys (T_send, T_rcv). Each direction has an independent 64-bit message counter; the receiver maintains a sliding anti-replay window (UNVERIFIED: exact window size — WireGuard uses a 2048-bit / 8000-packet-class window; pin against boringtun).

4.3 Device keys never leave the device (W3)

helix-wg generates the Curve25519 keypair on-device; only `local_public()` is exported for enrollment (doc 02 PKI) [01-DP §4, SYNTHESIS §7]. The private key is held in a SecretBytes zeroizing wrapper, never logged, never serialized to durable storage in plaintext (§14.1).

4.4 Post-quantum PSK seam (additive, Phase 2)

Post-quantum is an **additive pre-shared-key layer**, hybrid-never-PQ-only, that slots into the WG PSK field with **no transport change** [01-DP §4, SYNTHESIS §4]. Mechanism (parity target, grounded): establish a vanilla WG tunnel, run a PQ KEM exchange (ML-KEM-1024 / FIPS-203, optionally hybridized with Classic McEliece) *over* it, install the negotiated shared secret as WireGuard's PSK — the PSK is mixed into the symmetric key schedule so the tunnel stays secure even if X25519 is broken [research-mullvad §4]. `helix-wg`'s only surface is the `PeerConfig.psk: Option<Psk>` field (§2.2) + the key-schedule mix-in (§4.2); the full KEM exchange protocol is owned by the security doc, NOT here. Cost: ~1–2 s extra at handshake, steady-state unchanged [research-mullvad §4]. UNVERIFIED: the exact PQ-PSK message sequence — public Mullvad docs are high-level on this [research-mullvad §"Honest gaps"]; treat as security-doc design work.

5. AllowedIPs semantics — the crypto-routing trie

5.1 Two roles of one field (grounded)

AllowedIPs is WireGuard's **cryptographic routing table** and serves both directions [01-DP §7 layer 1, 04_ARCH §3.4]:

- **Outbound (routing):** for a plaintext packet leaving the TUN, the inner **destination** IP selects which peer to encrypt for — longest-prefix match across all peers' AllowedIPs.
- **Inbound (access control / W4):** for a packet decrypted from a peer, the inner **source** IP MUST fall inside *that* peer's AllowedIPs, else drop. A peer cannot spoof a source it was not granted — the coarse, crypto-enforced default-deny layer (I6); the port/proto-granular layer is the edge verdict map [01-DP §7 layer 2].

5.2 Data structure — longest-prefix-match trie

```
// helix-wg/src/allowed_ips.rs
/// A binary radix (Patricia) trie over IPv4+IPv6 prefixes →
///     PeerIndex,
/// matching the kernel WireGuard `allowedips` structure. O(W)
///     lookup
/// (W = address bit-width: 32 / 128), no allocation on lookup.
pub struct AllowedIpsTrie { /* dual v4/v6 radix roots */ }

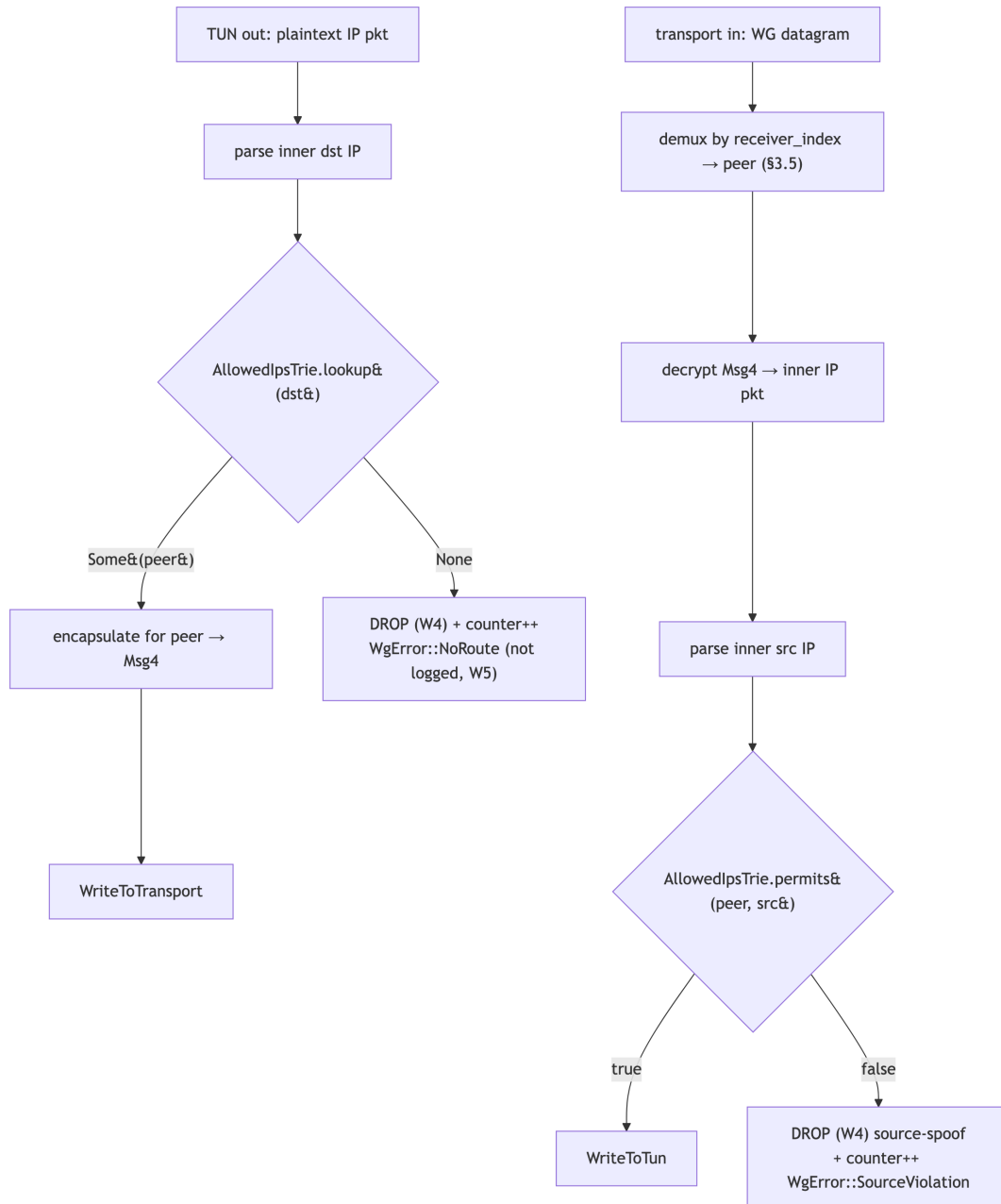
impl AllowedIpsTrie {
    pub fn insert(&mut self, prefix: IpNet, peer: PeerIndex);
    pub fn remove(&mut self, prefix: IpNet, peer: PeerIndex);
    pub fn remove_peer(&mut self, peer: PeerIndex); // revoke
        path (<1s, 01-DP §7)
    /// Longest-prefix match for an address → the peer that owns
    /// it (outbound routing).
    pub fn lookup(&self, addr: IpAddr) -> Option<PeerIndex>;
    /// True iff `addr` falls within ANY prefix granted to `peer`
    /// (inbound audit, W4).
    pub fn permits(&self, peer: PeerIndex, addr: IpAddr) -> bool;
}
```

5.3 Overlapping-prefix rule

Within one device, AllowedIPs prefixes across peers MUST be disjoint after longest-prefix resolution (WireGuard semantics: a more-specific prefix on peer B overrides a less-specific one on peer A). The RouteMap compiler (helix-route, [01-DP §6–§7]) guarantees disjointness *post-4via6* — the overlay ULA /48 + 4via6 mapping is exactly what prevents two connectors that both serve 192.168.1.0/24 from colliding here [01-DP §6.1, 04_ARCH

§3.4]. helix-wg trusts the compiled, disjoint overlay prefixes; it does **not** itself resolve RFC1918 collisions (that is solved one layer up). Edge case §12.4 covers a malformed (overlapping) map.

5.4 Crypto-routing decision flow



6. Per-device key generation + rotation

6.1 Generation (grounded, W3)

Each device (client, connector, edge) generates its Curve25519 keypair locally at enrollment; the private key never leaves the device, only the public key is registered with the control plane PKI [01-DP §4, 04_ARCH §7, research-mullvad §8].

```
// helix-wg/src/keys.rs
pub struct DeviceKeypair { pub public: PubKey, private:
    PrivKey /* zeroizing */ }
impl DeviceKeypair {
    /// Generate from a CSPRNG (OS getrandom). The ONLY way to
    /// mint a device key.
    pub fn generate() -> DeviceKeypair;
    /// Load a previously-generated private key from the platform
    /// secure store
    /// (Keychain / Keystore / TPM-backed) – never from a
    /// plaintext file (§14.1).
    pub fn from_secure_store(handle: SecureKeyHandle) ->
        Result<DeviceKeypair, WgError>;
    pub fn public(&self) -> PubKey;
}
```

6.2 Two rotation tiers (grounded)

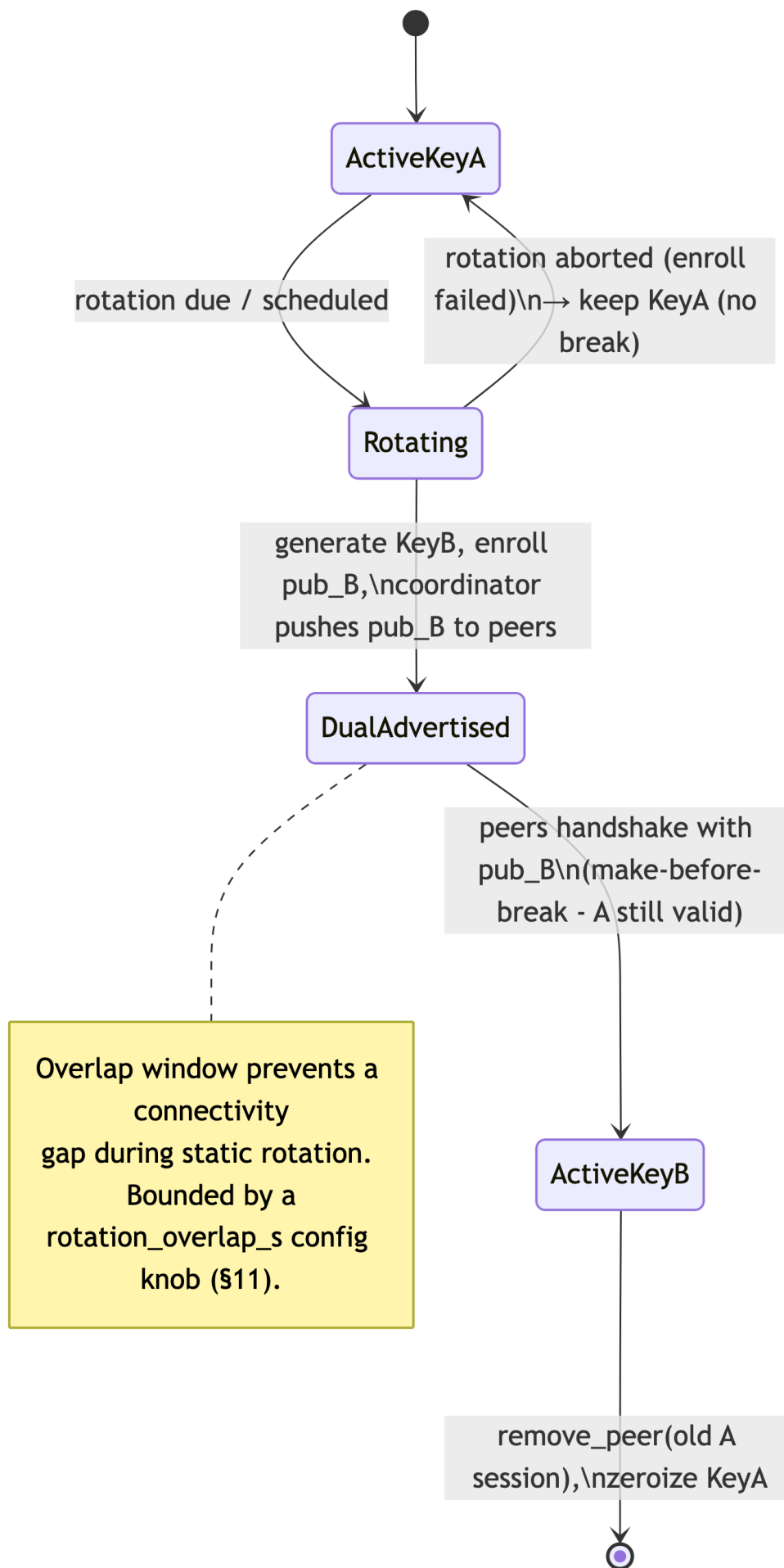
WireGuard distinguishes **ephemeral session rekey** (automatic, every ~2 min, §3.6 Rekeying) from **static device-key rotation** (policy-driven). Both exist [01-DP §4 "rotated on schedule and on device.revoked"; 04_ARCH §7].

Tier	Trigger	Mechanism	Owner
Session rekey	REKEY_AFTER_TIME (120s) /	new Noise IK handshake, same static keys (§3.6)	helix-wg (automatic)
	REKEY_AFTER_MESSAGES / inbound after		
Static rotation	REKEY_AFTER_TIME	mint new DeviceKeypair, re-enroll public key, coordinator pushes updated PeerRoute.wg_pubkey, peers upsert_peer the new pub, old peer remove_peer	helix-core + PKI (doc 02); helix-wg executes the upsert/remove
	schedule (e.g. 30/90d) OR device.revoked event		

6.3 Revocation is a peer removal (grounded, <1s)

device.revoked → the coordinator pushes a map delta removing the peer → helix-core calls `WgDevice::remove_peer` → the session is zeroized and the `AllowedIpsTrie` entry dropped, in < **1 second**, no restart [01-DP §7, 04_ARCH §4.4, SYNTHESIS §7]. On the edge, this composes with the verdict-map drop (layer 2); helix-wg owns the crypto-routing half.

6.4 Static-rotation state machine (make-before-break)



7. MTU derivation

7.1 The rule (grounded)

The inner WG MTU is derived from the **active transport's** `effective_mtu()` minus nothing further (the transport already subtracted its own overhead), then clamped by path-MTU discovery [01-DP §3.1 trait, §10]:

```
inner_wg_mtu = min( active_transport.effective_mtu(),
path_mtu_discovered )
```

`helix-wg` does **not** hardcode an MTU; it reads it from the transport it is handed (decoupling — the same WG core runs over any carrier, W6). The transport budget table is owned by [01-DP §10]; reproduced for the WG-overhead derivation:

Transport	<code>effective_mtu()</code>	WG-relevant overhead accounted	Source
plain-udp	1420	1500 – 20 IP – 8 UDP – 32 WG (16 hdr + 16 tag, §3.4)	[01-DP §10, 04_P0 §4.3]
masque-h3	1280 (measure & tune)	QUIC + HTTP-Datagram + UDP-proxy overhead; 1280 = IPv6 min-MTU floor	[01-DP §10, 04_P0 §5.2]
shadowsocks	~1380	TCP MSS – 2 B length prefix – AEAD tag	[01-DP §10, 04_P2 §1.1]
udp-over-tcp	~1380	TCP MSS – length prefix	[01-DP §10]
lwo	~1400	plain-udp – padding bytes	[01-DP §10]

7.2 The 32-byte WG overhead is already inside `effective_mtu()`

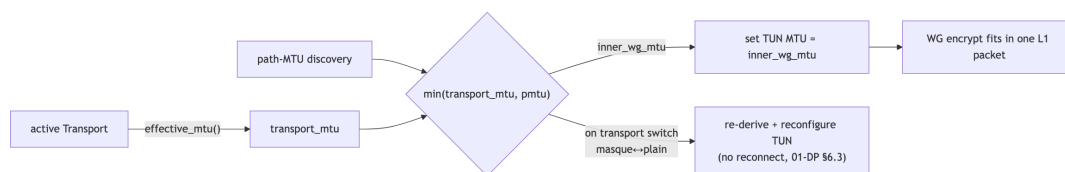
`plain-udp`'s 1420 already subtracts the WireGuard 16-byte data header + 16-byte Poly1305 tag (§3.4). So `helix-wg` configures the TUN MTU to exactly `inner_wg_mtu`; the transport's `effective_mtu()` is the post-WG-overhead figure for the *inner* IP payload. (UNVERIFIED: whether the 1420 figure folds

the WG overhead or the IP/UDP overhead — the pass-1 doc's arithmetic $1500 - 20 - 8 - 32 \approx 1420$ folds **both**; confirm the split during the §13 benchmark.)

7.3 Oversize handling

A plaintext packet whose encrypted size would exceed the active transport's MTU yields `TransportError::Oversize` at the transport boundary; `helix-wg`'s contract is to **lower the inner TUN MTU** (preferred) or signal L3 fragmentation rather than silently truncate [01-DP §10 rule 3]. The orchestrator sets the TUN MTU from `inner_wg_mtu`, so under normal operation oversize is unreachable; it is a defensive edge (§12.5). When DAITA pads (§9 of [01-DP]), padding is added **after** WG encrypt and counts against the transport budget, not the inner MTU [01-DP §10 rule 2] — `helix-wg` is unaware of DAITA padding (placement is L2.5, beneath WG output, above transport).

7.4 MTU derivation flow



When the ladder switches transport (e.g. `plain-udp` → `masque-h3`, [01-DP §5.3]), the inner MTU changes ($1420 \rightarrow 1280$) and `helix-core` re-derives + reconfigures the TUN MTU without a reconnect of unrelated state — a first-class reconcile event.

8. Timers, rekey, and keepalive state machine

8.1 The tick (grounded)

`helix-core` calls `WgDevice::tick(now, scratch)` every **TICK_INTERVAL** — $\sim 100\text{--}250$ ms [01-DP §4, 04_P0 §4.4 "every 100–250 ms"]. Each tick advances every peer's WireGuard timers (boringtun `update_timers`-equivalent, UNVERIFIED: API note 2) and returns the batch of `WgVerdicts` to emit (keepalives, rekey `Msg1s`, expirations).

8.2 Timer effects (grounded constants, §3.6)

Condition	Effect
REKEY_AFTER_TIME (120s) since last handshake, AND outbound traffic pending	initiator emits Msg1 (rekey)
KEEPALIVE_TIMEOUT (10s) since last <i>received</i> data, AND data was sent	passive keepalive (empty Msg4)
persistent_keepalive set (e.g. 25s)	unconditional keepalive every N seconds — keeps NAT mappings alive (§8.3)
REJECT_AFTER_TIME (180s) with no valid traffic	session expires → Down; next packet re-handshakes

8.3 Persistent keepalive (NAT survival)

PeerConfig.persistent_keepalive: Option<u16> (§2.2). The **connector** and **client** behind NAT set it (typical 25s) so the gateway-bound mapping never expires; the gateway edge generally does not (peers dial it). This composes with Phase-2 NAT traversal [01-DP §8] but is independent of it.

8.4 Roaming (endpoint latch — grounded)

WireGuard roaming: when a valid, authenticated Msg4 arrives from a *new* source SocketAddr, helix-wg updates WgPeer.endpoint to that address — no re-handshake [01-DP §8.2 "WireGuard's roaming latches onto whatever source address a valid handshake arrives from"]. This is what makes mobile network switches (Wi-Fi [?] cellular) seamless and is the substrate for Phase-2 hole punching [01-DP §8.2]. Only a *cryptographically valid* packet moves the endpoint — a spoofed source cannot hijack the session (§14.2).

9. The encrypted-datagram boundary (transport sits beneath)

9.1 The contract (grounded, I1/I2)

helix-wg [?] helix-transport exchange **only opaque, already-encrypted WG datagrams** as bytes::Bytes [01-DP §3.1, §0.1 I1/I2]. The transport never sees plaintext (I1) and carries unreliable datagrams, never an ordered stream (I2). The boundary is exactly two calls:

```

// helix-core orchestrator loop (01-DP §5.1) – the seam between §2
// and 01-DP §3:
// OUTBOUND
match device.handle_tun_out(&ip_pkt, &mut scratch) {
  WgVerdict::WriteToTransport(dg) =>
    transport.send(Bytes::copy_from_slice(dg)).await?,
  WgVerdict::Nothing => {},
  WgVerdict::Err(e) => metrics.wg_error(e),           // counter
    only (W5)
  WgVerdict::WriteToTun{..} => unreachable!(),       // tun_out
    never yields tun output
}
// INBOUND
let dg = transport.recv().await?;                   // opaque
  WG bytes (I1)
match device.handle_transport_in(&dg, from, &mut scratch) {
  WgVerdict::WriteToTun{ pkt, .. } => tun.write(pkt).await?,
  WgVerdict::WriteToTransport(reply) =>
    transport.send(Bytes::copy_from_slice(reply)).await?, //
    handshake reply
  WgVerdict::Nothing => {},
  WgVerdict::Err(e) => metrics.wg_error(e),
}

```

9.2 Why helix-wg is transport-agnostic (W6, decoupling)

helix-wg holds no reference to a concrete transport — it returns *bytes* and takes *bytes*. The orchestrator owns the `Box<dyn Transport>` and the ladder [01-DP §5.3]. This is what makes the *same* WG core run over plain-udp, masque-h3, shadowsocks, ... and what makes multi-hop trivial: an outer hop's Transport simply carries the inner hop's WG datagrams [01-DP §11.1]. helix-wg is unaware of obfuscation, DAITA, or which carrier is active — it only knows the active MTU (§7).

9.3 Kernel-backend boundary variation

When `WgBackend::LinuxKernel` runs plain-udp, the encrypt/decrypt happens in the kernel and the datagram boundary is a kernel UDP socket, not the `Transport::send/recv` calls (§1.3). For **any obfuscating transport**, the kernel cannot emit the carrier framing, so the userspace boundary (9.1) is mandatory — i.e. masque-h3/shadowsocks/... force `WgBackend::Userspace`. The `WgDataPath` trait (§1.3) hides this: helix-core calls the same verdict pump; the kernel impl short-circuits to netlink config + raw socket for plain-udp only. Edge case §12.7 enumerates the forbidden kernel+obfs combination.

9.4 Multi-hop nesting (grounded, Phase 2)

A 2-hop chain [Entry, Exit] is an ordered `Vec<Hop>` where `Hop = { wg: WgPeer, transport: Box<dyn Transport> }` [01-DP §11.1]. Outbound: encrypt for Exit (innermost WgPeer) → the Exit ciphertext is the *plaintext* fed to the Entry WgPeer's `handle_tun_out` → Entry ciphertext → `Entry.transport.send()`. `helix-wg` provides the per-hop WgPeer/WgDevice; the chain orchestration is `helix-core::hop_chain` [01-DP §11.1]. Entry sees the client but not the destination; Exit sees the destination but not the client — the Mullvad multi-hop property [01-DP §11.1, research-mullvad §3].

10. Error taxonomy

```
// helix-wg/src/error.rs
#[derive(thiserror::Error, Debug, Clone, PartialEq)]
pub enum WgError {
    // ---- handshake ----
    #[error("handshake timed out after {0:?}")]
    HandshakeTimeout(Duration), // REKEY_ATTEMPT_TIME
    exhausted
    #[error("handshake rejected: bad MAC1")]
    BadMac1, // anti-DoS first gate (§14.3)
    #[error("handshake rejected: bad MAC2/cookie")]
    BadMac2,
    #[error("handshake AEAD verification failed")]
    HandshakeAead, // wrong peer / tampered
    #[error("stale handshake timestamp (replay)")]
    StaleTimestamp, // TAI64N anti-replay (§3.2)
    // ---- transport-data ----
    #[error("AEAD decrypt failed (peer {0})")]
    DataAead(PeerIndex), // tampered / wrong key
    #[error("replayed counter (peer {0})")]
    Replay(PeerIndex), // anti-replay window (§4.2)
    #[error("unknown receiver index {0}")]
    UnknownIndex(u32), // demux miss (§3.5)
    #[error("malformed WG datagram (len {0})")]
    Malformed(usize),
    // ---- routing / access control (W4) ----
    #[error("no AllowedIPs route for dst {0}")]
    NoRoute(IpAddr), // outbound drop (§5)
    #[error("source {src} not permitted for peer {peer}")]
    SourceViolation { peer: PeerIndex, src: IpAddr },
    // ---- session lifecycle ----
    #[error("no current session (peer {0}); handshake required")]
    NoSession(PeerIndex),
    #[error("peer {0} expired (REJECT_AFTER_TIME)")]
    SessionExpired(PeerIndex),
    // ---- config / key ----
    #[error("invalid key material")]
    InvalidKey,
```

```

    #[error("secure store error: {0}")]
    SecureStore(String),
}

```

Classification rules (anti-bluff, §11.4.1/.6):

- NoRoute / SourceViolation / Replay / BadMac1 are **expected, policy-or-attack-driven drops** — counted (PeerStats), surfaced as metrics, never an Err propagated to the user as a tunnel failure, never logged per-packet (W5). They map to WgVerdict::Nothing at the orchestrator with a counter bump, NOT a Down{reason} status.
 - HandshakeTimeout / HandshakeAead are **genuine connection failures** → surface as TunnelStatus::Reconnecting then ladder escalation [01-DP §5.3].
 - Malformed / UnknownIndex are **defensive drops** at the edge under scan load — bounded by the rate limiter (§14.3), counted, never logged.
 - A script-internal failure (panic, OOB slice) is a §11.4.1 FAIL-bluff and MUST fail the build, never reach this enum.
-

11. Configuration knobs

Only **operational** knobs are exposed; crypto parameters (§4) and WireGuard timer constants (§3.6) are **fixed** (W2) and are read-only constants, NOT config.

```

// helix-wg/src/config.rs
#[derive(Clone, Debug)]
pub struct WgDeviceConfig {
    pub prefer_kernel: bool,           // Linux only; resolve()
    honours_availability (§1.3)
    pub tick_interval: Duration,       // default 200ms; clamp
    [100ms, 250ms] (§8.1)
    pub rotation_overlap_s: u32,       // static-key make-before-
    break window (§6.4); default 120
    pub handshake_rate_limit: RateLimit, // per-source cookie-
    challenge threshold (§14.3)
    pub scratch_size:
        usize,           // per-loop scratch buffer bytes;
        default 65535 (max IP pkt)
    pub max_peers: u32,           // edge guardrail; client/
        connector = 1
}
// Per-peer knobs live in PeerConfig (§2.2): allowed_ips,
// endpoint,
// persistent_keepalive, psk. All are reconciler-pushed from the
// RouteMap (01-DP §6.2).

```

Defaults are chosen so the client/connector single-peer case needs **zero** explicit config; the edge sets max_peers + a tighter handshake_rate_limit.

12. Edge cases (enumerate, do not hand-wave — §11.4.118)

#	Edge case	Required behaviour
12.1	Inbound Msg4 before any handshake completes	WgError::NoSession → drop + counter; do NOT emit plaintext. Trigger a rekey if outbound traffic is queued.
12.2	Two concurrent handshake initiations (both sides initiate)	WireGuard tie-break by handshake hash / timestamp; keep the winner, drop the loser. UNVERIFIED: exact tie-break — pin to boringtun.
12.3	Peer removed (remove_peer) mid-decrypt of its datagram	Datagram in flight is dropped (UnknownIndex); zeroize must complete before the index is reused.
12.4	Malformed/overlapping AllowedIPs in a pushed map	helix-wg trusts the compiler's disjointness (§5.3); a longest-prefix collision resolves deterministically; an <i>invalid</i> prefix is rejected at upsert_peer with InvalidKey-class error, map delta NACKed to helix-core.
12.5	Plaintext packet larger than inner_wg_mtu	Lower TUN MTU (preferred) or L3-fragment; never silent truncate (§7.3).
12.6	Transport switch (ladder) mid-session	Re-derive MTU (§7.4); the WG session is preserved across transport change — same keys, new carrier; no re-handshake unless the endpoint changed.
12.7	WgBackend::LinuxKernel requested with an obfuscating transport	Forbidden combination → force Userspace for that tunnel (§9.3); surface a one-time warning, never a silent wrong-backend run (§11.4.6).
12.8	Counter exhaustion (REJECT_AFTER_MESSAGES)	Force rekey before the 64-bit counter nears exhaustion; a session that cannot rekey expires (SessionExpired).
12.9	Clock skew breaking TAI64N timestamp check	StaleTimestamp drop; the device clock MUST be monotonic-enough

#	Edge case	Required behaviour
		for handshake — surface a health warning, do not weaken the check.
12.10	iOS NE memory pressure during QUIC buffering	helix-wg itself is alloc-free on the hot path (W1); buffering lives in the transport. The memory gate G3 is a <i>transport+core</i> measurement, not a helix-wg-only one [04_P0 §6].
12.11	Roaming: valid Msg4 from a new source while old endpoint still sends	Latch to the newest <i>cryptographically valid</i> source (§8.4); ignore the stale one.

13. Test points (§11.4.169-class; types per §11.4.27 — see note 3)

Every claim ships **captured evidence**, not a config-only PASS (§11.4.5/.69/.107). UNVERIFIED: §11.4.169's exact enumeration (note 3); the types below are the in-context §11.4.27 set.

Test type	What it exercises	Evidence (anti-bluff)	Maps to gate
unit	AllowedIpsTrie longest-prefix + permits; WgVerdict mapping from TunnResult; MTU derivation arithmetic (§7.1); error classification (§10)	property tests: random prefix sets vs a reference linear matcher; assert byte-identical routing decisions	—
fuzz	handle_transport_in on arbitrary bytes (malformed datagrams, §3.5/§12)	cargo-fuzz corpus; zero panics (W1/ §11.4.1), every input → a WgVerdict (drop or output), never a crash	—

Test type	What it exercises	Evidence (anti-bluff)	Maps to gate
integration	client [?] gateway [?] connector plain-UDP WG up; ping overlay IP; reach 10.10.0.20:80 through the netns rig	iperf3 ≥ 80% bare link + curl hello page CSV from bench.sh	G1 / S1–S2 [04_P0 §8]
integration (obfs)	same slice over masque-h3 with plain WG blocked (nft rule)	tshark capture classified HTTP/3, no WG signature ; goodput @ 5% netem loss beats UoT strawman	G2 / S3 [04_P0 §5.3, §8]
e2e	full ladder escalation plain- udp → ... → masque-h3; session survives transport switch (§12.6)	ordered TunnelStatus event trace captured; WG session keys unchanged across switch	— [01- DP §5.4]
security	source-spoof drop (W4/§5.4); replay drop (§4.2); MAC1/ cookie anti-DoS (§14.3); revoked peer unreachable in <1s (§6.3)	denied-flow packet capture; revoke→drop timing < 1s captured	[01-DP §7]
performance / benchmark	encrypt/decrypt throughput; CPU per Gbps; handshake/sec churn; per-platform RSS (W1)	bench.sh CSV: throughput @ 1/10/100 peers, p50/ p99, CPU/ Gbps; iOS Instruments RSS soak	G3 (iOS mem), G4 (edge) [04_P0 §6–§8]
stress / chaos (§11.4.85)	sustained N≥100-iteration handshake churn; mid- handshake peer removal (§12.3); clock-skew injection	per-iteration latency.json + categorised- error census;	—

Test type	What it exercises	Evidence (anti-bluff)	Maps to gate
	(§12.9); counter-exhaustion forced rekey (§12.8)	recovery trace; no deadlock/leak	
deterministic-consistency (§11.4.50)	the integration + obfs slices run N times (3 normal / 10 cycle) on the same rig	identical PASS + identical evidence hashes every run	—
Challenge / HelixQA (§11.4.27)	the §11.4.169-class per-feature "really works" challenge: drive the real client→LAN journey, confirm the hello page, captured	HelixQA result.json + captured transcript under docs/qa/<run-id>/ (§11.4.83)	—

Each fix to helix-wg follows §11.4.115 RED-on-broken + §11.4.135 standing regression guard (e.g. the source-spoof-accept bug → a permanent permits()-rejects-spoof guard).

14. Security considerations

- 1. Private key custody (W3, §4.3/§6.1).** Device private keys live in the platform secure store (iOS Keychain / Android Keystore / TPM-backed where available), SecretBytes-wrapped (zeroizing), never logged, never written to a plaintext file [01-DP §4, 04_ARCH §7, research-mullvad §8]. A leak path is a §11.4.10 violation.
- 2. Cryptographic source authentication (§8.4/§12.11).** Only a valid AEAD- authenticated Msg4 moves a peer's latched endpoint; a spoofed-source packet cannot hijack a session or redirect traffic — roaming is safe by construction.
- 3. Handshake anti-DoS (§14.3, grounded architecture).** MAC1 (keyed BLAKE2s over the static pubkey) is the cheap first gate: a Msg1 with a bad MAC1 is dropped before any expensive Curve25519 work. Under load the responder issues a **Cookie Reply (Msg3, §3.1)** demanding MAC2, rate-limited per source — the edge's defence against handshake floods. UNVERIFIED: exact cookie TTL / rate-limit constants (pin to WireGuard whitepaper + boringtun).
- 4. No-logging by construction (W5).** Only aggregate PeerStats counters exist; no per-packet, per-connection, or durable session state. The edge's

sim-connection accounting is in-RAM only [research-mullvad §8, 01-DP §0.1 I5].

5. **Default-deny crypto-routing (W4/§5).** A peer reaches only its granted AllowedIPs; this is the coarse crypto layer beneath the port/proto edge verdict map — both required (defense in depth) [01-DP §7].
6. **Harvest-now-decrypt-later (§4.4).** The additive PQ-PSK seam protects today's recorded traffic against future quantum decryption, hybrid (never PQ-only) so a flaw in either KEM alone does not break the PSK [research-mullvad §4].
7. **Target-system safety (§11.4.133).** The Linux kernel-WG netlink path writes only the documented WireGuard config surface; no out-of-range or unverified kernel writes. Backend selection never bricks the host.

15. Performance budget

Metric	Target	Source
plain-UDP through-tunnel throughput	≥ 80% of bare link	G1 [04_P0 §8]
masque-h3 through-tunnel throughput	≥ 50% of plain-UDP	G2 [04_P0 §8]
Added latency (plain-UDP)	< 2 ms	[04_P0 §8]
Added latency (masque-h3)	< 15 ms	[04_P0 §8]
Handshake time (plain)	< 1 s	[04_P0 §8]
Handshake time (masque-h3)	< 2 s	[04_P0 §8]
Reconnect / roam recovery	< 3 s	[04_P0 §8]
iOS NE steady-state RSS (core, both transports)	under device ceiling with ≥ 30% headroom , 30-min soak	G3 (make-or-break) [04_P0 §6.3]
Steady-state heap alloc on encrypt/decrypt	zero (scratch-buffer only)	W1
Edge CPU per Gbps	recorded (cost-to-serve, feeds D5 edge-language decision)	G4 [04_P0 §7–§8]
PQ-PSK handshake extra cost	~1–2 s at handshake; steady-state unchanged	research-mullvad §4

The MASQUE throughput penalty (double crypto + double congestion control) is **real and must be measured, never claimed at parity** [research-masque §5 "do NOT claim parity speed"]. The §13 benchmark fills these cells with captured numbers (§11.4.5/.69/.123).

16. Build checklist (this crate)

File	Owns	Phase	Gate
helix-wg/src/lib.rs	WgVerdict, re-exports	0	G1 / S1
helix-wg/src/backend.rs	WgBackend, WgDataPath, resolve() (§1.3)	0→1	G1
helix-wg/src/peer.rs	WgPeer, WgDevice, pump methods (§2)	0	G1 / S1
helix-wg/src/allowed_ips.rs	AllowedIpsTrie (§5)	0→1	G6 / S8
helix-wg/src/keys.rs	DeviceKeypair, rotation (§6)	0→1	—
helix-wg/src/config.rs	WgDeviceConfig (§11)	0	—
helix-wg/src/error.rs	WgError taxonomy (§10)	0	S0

Surviving interfaces (frozen contracts per [01-DP §4, 04_P0 §14]): WgVerdict, WgPeer/WgDevice pump signatures, the encrypted-datagram boundary (§9). Their implementations may evolve; the signatures hold from Phase 0.

End of helix-wg nano-detail specification (Volume 2, doc 01). Pair with [01-DP] §4 (the section this deepens), the Transport trait spec ([01-DP] §3 — the layer beneath this one at the §9 boundary), and the security doc (PQ-PSK KEM exchange §4.4, kill-switch/DNS-leak state machine). Wire-byte layouts (§3) and boringtun signatures (§2) are UNVERIFIED: against the evidence base and MUST be confirmed against the WireGuard whitepaper + the pinned boringtun crate before implementation (§11.4.6).